

1. **Download any free APK file (for example, a simple calculator app) and install it on your Android emulator using Android Studio's Device Manager.**

ANS:

- Downloaded a free Notes APK application.
- Installed the APK successfully on the Android Emulator.
- Installation process completed smoothly without any issues.
- The application was visible on the emulator home screen after installation.

2. **Open the installed app on your emulator, rotate the device from portrait to landscape and back, and note any UI issues or layout problems that appear.**
Hint: Use the emulator's rotate button to change orientation and take screenshots if you spot any glitches.

ANS:

- Opened the installed Notes application on the emulator.
- Changed the screen orientation from Portrait to Landscape mode.
- Switched back from Landscape to Portrait mode.
- The application adjusted correctly to both screen orientations.
- No UI misalignment or content overlapping was observed.

3. **Simulate turning off the internet connection in your emulator, then try to use a feature in the app that requires network (like fetching data or logging in). Record the error message or behavior you observe.**

ANS:

- Disabled the internet connection on the emulator.
- Opened the application and attempted to access online content.
- The application displayed a message indicating that the network was unavailable.
- Once the internet connection was enabled again, the application resumed normal operation.

4. **Test the login or registration flow of any installed app on both your emulator and a real Android phone (if available). Compare the experience and list at least two differences you notice in speed, UI, or notifications.**

ANS:

- Tested the application on both an Android Emulator and a real Android phone.
- The application launched more quickly on the real device.
- Scrolling and touch responses were smoother on the physical device.
- Notifications were received more consistently on the real device than on the emulator.
- Overall functionality was similar on both platforms.

5. Pick any app you use daily (Instagram, Zomato, Paytm, etc.) and describe how you would test push notifications using both an emulator and a real device. Mention at least one limitation of emulators for this test.

ANS:

- Chose Paytm for push notification testing.
- Confirmed that notification permissions were enabled.
- Triggered notifications by performing account-related activities.
- Notifications were received successfully on the real device.
- Notification delivery on the emulator was slightly slower.
- Limitation: Emulators may not fully support background notification services and real-time notification delivery.